

Miriam A. Janssen

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Education & Appointments

Dartmouth College, Department of Psychological and Brain Sciences, Hanover, NH Fall 2022 - present

Ph.D. Psychological and Brain Sciences

M.A. Psychological and Brain Sciences

MVDM laboratory

Advisor: Matthijs A. A. van der Meer, PhD

Thesis: Investigating Valenced Replay and Teaching Signals

National Institutes of Health, The National Institute of Mental Health, Bethesda, MD Fall 2020 - Spring 2022

Postbaccalaureate intramural research training award fellow

Section on Learning and Decision Making

Advisor: Bruno B. Averbeck, PhD

Projects: (1) The effects of periaqueductal gray lesions in rhesus monkeys on attention through a free view task, (2) The role of the perirhinal cortex, ventral striatum, and amygdala in learning from gains and losses on a conditioned reinforcement task, (3) Hierarchical reinforcement learning and the dorsal frontostriatal system

University of Maryland, College of Behavioral & Social Sciences, College Park, MD Spring 2021

University of Maryland, A. James Clark School of Engineering, College Park, MD Spring 2020

B.S., Bioengineering; B.A., Economics

Research Assistant

Tissue Engineering & Biomaterials Laboratory

Advisors: John P. Fisher, PhD & Justine R. Yu, MD/PhD

Project: Gelatin Methacrylate Hydrogel Delivery System for Stimulation of Mesenchymal Stem Cell Migration

Awards & Fellowships

E.E. Just Graduate Fellowship Program, Dartmouth College | \$21,600* 2023-present

Marie A. Center 1982 Award for Excellence in Teaching, Dartmouth College, PBS department 2025

Course: Systems Neuroscience

D. G. Marquis Behavioral Neuroscience Award, American Psychological Association, Division 6 2024

Paper title: Taswell, C. A., Janssen, M., Murray, E. A., & Averbeck, B. B. (2023). The motivational role of the ventral striatum and amygdala in learning from gains and losses. *Behavioral Neuroscience*, 137(4), 268–280.

<https://doi.org/10.1037/bne0000558>

National Science Foundation Graduate Research Fellowship Program | \$138,000** 2023-2026

National Institutes of Health Intramural Research Training Award 2020-2022

Fischell Department of Bioengineering Capstone Design Competition - 3rd Place 2019

Project: Reusable, robust, and non-electronic autoclave verification device for austere environments

Partner: Engineering World Health

Dean's List, University of Maryland 2019-2020

Dean's List, University of Maryland 2016-2017

*Including only direct costs; **Including both direct and indirect costs

Publications

Janssen M.A., Chen H.T., Tritsch N.X., van der Meer M.A.A. (2025) Ventral Striatal Dopamine Increases following Hippocampal Sharp-Wave Ripples. *bioRxiv* [Preprint]. doi: 10.1101/2025.07.24.666687. PMID: 40766361; PMCID: PMC12324199.

- Analysis repository: <https://github.com/mimijanssen/replayDA>
- Curated dataset: [Link](#)

Taswell, C.A., **Janssen, M.**, Murray, E.A., & Averbeck, B.B. (2023). The Motivational Role of the Ventral Striatum and Amygdala in Learning from Gains and Losses. *Behavioral Neuroscience*, 137 (4): 268–280.

Janssen, M.*, LeWarne, C.*, Burk, D., & Averbeck, B.B. (2022). Hierarchical reinforcement learning, sequential behavior and the dorsal frontostriatal system. *Journal of Cognitive Neuroscience*, 34 (8): 1307–1325.

Justine, R.Y., **Janssen, M.**, Liang, B.J., Huang, H.C., & Fisher, J.P. (2020). A liposome/gelatin methacrylate nanocomposite hydrogel system for delivery of stromal cell-derived factor-1 α and stimulation of cell migration. *Acta Biomaterialia*, 108, 67-76.

*Denotes equal contribution; underline indicates trainee

Talks & Presentations

- Gibbens, E.*, **Janssen, M.A.**, Sun, J., van der Meer, M.A.A., (2026). Encoding and reactivation of odor cues and their associated valenced outcomes along the dorsoventral hippocampal. *Poster at the Society for Neuroscience Annual Meeting*, Washington, DC.
- **Janssen, M.A.**, Chen, H.T., van der Meer, M.A.A., (2026). Ventral Striatal Dopamine Increases following dCA1 Sharp Wave-Ripples during Rest but not during Active Task Engagement. *Poster at the Neurobiology of Cognition, Gordon Research Seminar & Conference*, Waterville Valley, NH.
- Gibbens, E.*, **Janssen, M.A.**, Sun, J., van der Meer, M.A.A., (2026). Encoding and reactivation of odor cues and associated appetitive or aversive outcomes along the dorsoventral hippocampal. *Talk at the Vermont Summer Summit*, Burlington, VT.
- **Janssen, M.A.**, Chen, H.T., van der Meer, M.A.A., (2025). Ventral Striatal Dopamine Increases following Hippocampal Sharp-Wave Ripples. *Talk at the Vermont Summer Summit*, Burlington, VT.
- **Janssen, M.A.**, Chen, H.T., van der Meer, M.A.A., (2024). The Relationship Between Hippocampal Sharp Wave-Ripples and Ventral Striatal Dopamine. *Poster at the Society for Neuroscience Annual Meeting*, Chicago, IL.
- Taswell, C. A., **Janssen, M.**, & Averbeck, B.B., (2022). Differential value coding of learning from gains and losses in multiple prefrontal areas. *Poster at the Society for Neuroscience Annual Meeting*, San Diego, CA.
- **Janssen, M.A.**, Taswell, C. A., & Averbeck, B.B., (2022). The role of the ventral striatum and amygdala in learning from gains and losses. *Poster at the Julius Axelrod Symposium*, Bethesda, MD (online).
- **Janssen, M.A.**, Taswell, C. A., & Averbeck, B.B., (2021). The ventral striatum and amygdala's role in learning from gains and losses. *Talk at the Fellows Afternoon Neuroscience Seminars*, Bethesda, MD (online).
- **Janssen, M.A.**, Taswell, C. A., & Averbeck, B.B., (2021). The amygdala's role in learning from gains and losses. *Poster at the Society for Neuroscience Annual Meeting*, Chicago, IL (online).
- **Janssen, M.A.***, LeWarne, C.* & Averbeck, B.B., (2021). Hierarchical reinforcement learning and execution of motor sequences: a review. *Poster at the NIH Postbac Poster Day*, Bethesda, MD (online).
- Justine, R.Y., **Janssen, M.A.**, Liang, B.J., Huang, H.C., & Fisher, J.P., (2019). A Liposome/Gelatin Methacrylate Nanocomposite Hydrogel System For SDF-1 α Delivery And Stimulation Of Cell Migration. *Poster at the Biomedical Engineering Society*, Philadelphia, PA.
- **Janssen, M.A.**, Justine, R.Y., & Fisher, J.P., (2019). Gelatin Methacrylate Hydrogel Delivery System for Stimulation of Mesenchymal Stem Cell Migration. *Poster at the Biomedical Engineering Society*, Philadelphia, PA.

*Denotes presenter; underline indicates trainee

Department Talks

- April 14, 2026: Behavioral Brown Bag Talk Series at Dartmouth College, NH. "Valence Coupled Replay and Teaching Signals".
- September 23, 2025: Behavioral Brown Bag Talk Series at Dartmouth College, NH. Journal Club on the paper: Striatal Dopamine Signals Errors in Prediction Across Different Informational Domains
- February 18, 2025: Behavioral Brown Bag Talk Series at Dartmouth College, NH. Journal Club on the paper: Spatial localization of hippocampal replay requires dopamine signaling.
- April 23, 2024: Behavioral Brown Bag Talk Series at Dartmouth College, NH. "The Interaction between Hippocampal Sharp Wave Ripples and Ventral Striatal Dopamine".
- October 17, 2023: Behavioral Brown Bag Talk Series at Dartmouth College, NH. Journal Club on the paper: Revealing abrupt transitions from goal-directed to habitual behavior.
- May 9, 2023: Behavioral Brown Bag Talk Series at Dartmouth College, NH. Journal Club on the paper: Dopaminergic prediction errors in the ventral tegmental area reflect a multithreaded predictive model.
- January 31, 2023: Behavioral Brown Bag Talk Series at Dartmouth College, NH. "The Interaction between Hippocampal Sharp Wave Ripples and Ventral Striatal Dopamine".

Teaching Experience

Teaching Fellow , Psychological and Brain Sciences Department, Dartmouth College, NH Behavioral Neuroscience	Winter 2026
Teaching Fellow , Psychological and Brain Sciences Department, Dartmouth College, NH Systems Neuroscience Laboratory	Spring 2025
Teaching Fellow , Psychological and Brain Sciences Department, Dartmouth College, NH Systems Neuroscience Laboratory	Fall 2024
Teaching Assistant , Marine Biological Laboratory, University of Chicago, Woods Hole, MA Neural Systems and Behavior	Summer 2024
Teaching Fellow , Psychological and Brain Sciences Department, Dartmouth College, NH Introduction to Neuroscience Guest Lecture: Memory Systems	Winter 2024
Teaching Assistant , Chemistry Department, University of Maryland, College Park, MD General Chemistry Laboratory	Fall 2017

Additional Research Training

Computational Neuroscience , Neuromatch Academy	Summer 2023
Machine Learning with Python , Free Code Camp	Spring 2022
Methods in Neuroscience Course , National Institutes of Health	Spring 2022
The Biomethodology of the Laboratory Mouse , Office of Animal Care and Use, National Institutes of Health.	April 2022

Mentorship Experience

Full-time Research Staff

2025-present	Emma Gibbens Research Assistant/Lab Manager Project: Encoding and Reactivation of Odor Cues and their Associated Valenced Outcomes along the Dorsoventral Hippocampal Axis
2025-2026	Jing-tang Sun Research Assistant/Lab Manager Project: Odor Onset and Offset Coding in Mouse Hippocampal Neurons <i>Now: Research Assistant, UC Irvine (PI: Ron Frostig)</i>
2024-2025	Kyoko Leaman Research Assistant/Lab Manager Project: Reactivation of Discrete Odor Stimuli in the Mouse <i>Now: PhD student, Brown Neuroscience (PI: Anda Chirila)</i>

Undergraduate Thesis Students

2026-present	Annabelle Tolth Neuroscience Project: The Role of Theta Oscillations in Encoding and Reactivation of Valenced Odor Cues
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Undergraduate Research Assistants

2026	Molly Mendoza Neuroscience
2025-2026	Ellie Oyeboode Psychology Project: Temporal Coupling Between Dopamine Transients and Dentate Spikes During Offline Rest
2022-2023	Namitha Alluri Women in Science Program <i>Now: Medical Student, Yale School of Medicine</i>

E.E. Just STEM Community at Dartmouth College

Fall 2023-present

- Near-peer mentor for undergraduates in the E.E. Just STEM community, a STEM ecosystem for systemically excluded and racialized minorities. Since my sustained mentorship, the community has increased in participation by 60%.
- Led a professional development series covering parts of the graduate school application and workshopping student applications and grant applications.

Academic Service

Graduate Representative, Psychological and Brain Science Department, Dartmouth College Fall 2025-Summer 2026

Graduate Student Council Member, Committee for Addressing Racism and Equity, Dartmouth College Fall 2025

Co-organizer, Behavioral Brown Bag Talk Series, Psychological and Brain Science Department, Dartmouth College. 2023

Science Communication & Public Engagement

K-12 Outreach | Various Programs, Upper Valley, NH & VT 2025-present

- Taught sheep brain dissections and human brain anatomy for:
 - E.E. Just Dartmouth Adventures in STEM preorientation program (2025)
 - Dartmouth's Discovery Program for high schoolers (2025)
 - Dartmouth's Science day for the Upper Valley broader community (2025)
 - Kingdom Career Connect for 7th and 8th grade students in the rural Vermont Northeast Kingdom (2025 & 2026)
- Presented a poster on neuroscience research and careers in neuroscience for:
 - Dartmouth's Science day with Semillas to Science (2026)

New Hampshire Union Leader, Manchester, NH 2026

- Published an Op-Ed in the highest-read New Hampshire newspaper on "Why Science Funding Matters" as part of the Cornell McClintock Letters Initiative

Court Appointed Special Advocates Guardians ad Litem, Prince George's County, MD Spring 2019 – Fall 2021

CASA Volunteer

- Engaged with a foster child and those around them to identify needs and advocated for services in court

Branchville Volunteer Fire Company, Prince George's County, MD

June 2017 – Summer 2019

Emergency Medical Technician

- Treated and transported patients to hospitals in the most active combination fire department in the U.S.
- Volunteered over 1,000 hours.